

Lab 1 – Use-Case Flow Specification

Problem Statement #41: OpenAPI Mock Server Generator

Core Use Case: UC-02 – Generate Mock Endpoints

Primary Actor: API Developer

This use case describes generation of dynamic HTTP mock endpoints from an OpenAPI 3.0 YAML/JSON specification.

1. Preconditions

- The OpenAPI Mock Server Generator is available to the API Developer.
- The API Developer has an OpenAPI 3.0 specification in YAML or JSON format.
- The specification contains API paths/operations and relevant schema definitions.

2. Main Success Scenario

1. The API Developer selects the option to import an OpenAPI specification.
2. The system accepts the OpenAPI 3.0 YAML/JSON specification.
3. The system parses the specification and identifies API paths, operations, parameters, and schema definitions.
4. The system validates that the specification can be used to create mock endpoints.
5. The system generates a dynamic HTTP mock endpoint for each defined operation.
6. The system associates each generated endpoint with the corresponding request and response schemas.
7. The system configures randomized schema-compliant JSON responses for the generated endpoints.
8. The system activates the generated mock endpoints.
9. The system displays the generated HTTP methods and paths to the API Developer.
10. The use case ends successfully.

3. Alternate Flow – Invalid OpenAPI Specification

- 3a. The system detects that the imported specification is invalid or cannot be parsed.
 - 3a1. The system stops endpoint generation.
 - 3a2. The system displays a validation/error message identifying the problem.
 - 3a3. The API Developer corrects the specification and submits it again.
 - 3a4. The system resumes the main success scenario from Step 2.

4. Postconditions

- Mock endpoints corresponding to the valid operations in the specification are generated and activated.
- Generated responses are configured to follow the relevant response schemas.
- The generated endpoint list is available to the API Developer for use in testing.